Locally compact topological vector spaces

Terence Tao · 24 May 2011

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Recall that a (real) topological vector space is a real vector space $V = (V, 0, +, \cdot)$ equipped with a topology $\mathcal{F}$ that makes the vector space operations $+ : V \times V \to V$ and $\cdot : \mathbf{R} \times V \to V$ continuous. One often restricts attention to Hausdorff topological vector spaces; in practice, this is not a severe restriction because it turns out that any topological vector space can be made Hausdorff by quotienting out the closure $\overline{\{0\}}$ of the origin $\{0\}$. One can also discuss complex topological vector spaces, and the theory is not significantly different; but for sake of exposition we shall restrict attention here to the real case.

An obvious example of a topological vector space is a finite-dimensional vector space such as $\mathbf{R}^n$ with the usual topology. Of course, there are plenty of infinite-dimensional topological vector spaces also, such as infinite-dimensional normed vector spaces (with the strong, weak, or weak-* topologies) or Frechet spaces.

One way to distinguish the finite and infinite dimensional topological vector spaces is via local compactness. Recall that a topological space is locally compact if every point in that space has a compact neighbourhood. From the Heine-Borel theorem, all finite-dimensional vector spaces (with the usual topology) are locally compact. In infinite dimensions, one can trivially make a vector space locally compact by giving it a trivial topology, but once one restricts to the Hausdorff case, it seems impossible to make a space locally compact. For instance, in an infinite-dimensional normed vector space V with the strong topology, an iteration of the Riesz lemma shows that the closed unit ball B in that space contains an infinite sequence with no convergent subsequence, which (by the Heine-Borel theorem) implies that V cannot be locally compact. If one gives V the weak-* topology instead, then B is now compact by the Banach-Alaoglu theorem, but is no longer a neighbourhood of the identity in this topology. In fact, we have the following result:

> **Theorem 1** Every locally compact Hausdorff topological vector space is finite-dimensional.

The first proof of this theorem that I am aware of is by André Weil. There is also a related result:

> **Theorem 2** Every finite-dimensional Hausdorff topological vector space has the usual topology.

As a corollary, every locally compact Hausdorff topological vector space is in fact isomorphic to $\mathbf{R}^n$ with the usual topology for some n. This can be viewed as a very special case of the theorem of Gleason, which is a key component of the solution to Hilbert's fifth problem, that a locally compact group G with *no small subgroups* (in the sense that there is a neighbourhood of the identity that contains no non-trivial subgroups) is necessarily isomorphic to a Lie group. Indeed, Theorem 1 is in fact used in the proof of Gleason's theorem (the rough idea being to first locate a "tangent space" to G at the origin, with the tangent vectors described by "one-parameter subgroups" of G, and show that this space is a locally compact Hausdorff topological space, and hence finite dimensional by Theorem 1).

Theorem 2 may seem devoid of content, but it does contain some subtleties, as it hinges crucially on the *joint* continuity of the vector space operations $+ : V \times V \to V$ and $\cdot : \mathbf{R} \times V \to V$, and not just on the separate continuity in each coordinate. Consider for instance the one-dimensional vector space $\mathbf{R}$ with the *co-compact* topology (a non-empty set is open iff its complement is compact in the usual topology). In this topology, the space is T_1 (though not Hausdorff), the scalar multiplication map $\cdot : \mathbf{R} \times \mathbf{R} \to \mathbf{R}$ is jointly continuous as long as the scalar is not zero, and the addition map $+ : \mathbf{R} \times \mathbf{R} \to \mathbf{R}$ is continuous in each coordinate (i.e. translations are continuous), but not jointly continuous; for instance,

the set $\{(x,y) \in \mathbf{R} : x+y \notin [0,1]\}$ does not contain a non-trivial Cartesian product of two sets that are open in the co-compact topology. So this is not a counterexample to Theorem 2. Similarly for the cocountable or cofinite topologies on $\mathbf{R}$ (the latter topology, incidentally, is the same as the Zariski topology on $\mathbf{R}$).

Another near-counterexample comes from the topology of $\mathbf{R}$ inherited by pulling back the usual topology on the unit circle $\mathbf{R}/\mathbf{Z}$. Admittedly, this pullback topology is not quite Hausdorff, but the addition map $+ : \mathbf{R} \times \mathbf{R} \to \mathbf{R}$ is jointly continuous. On the other hand, the scalar multiplication map $\cdot : \mathbf{R} \times \mathbf{R} \to \mathbf{R}$ is not continuous at all. A slight variant of this topology comes from pulling back the usual topology on the torus $(\mathbf{R}/\mathbf{Z})^2$ under the map $x \mapsto (x, \alpha x)$ for some irrational α; this restores the Hausdorff property, and addition is still jointly continuous, but multiplication remains discontinuous.

As some final examples, consider $\mathbf{R}$ with the discrete topology; here, the topology is Hausdorff, addition is jointly continuous, and every dilation is continuous, but multiplication is not jointly continuous. If one instead gives $\mathbf{R}$ the half-open topology, then again the topology is Hausdorff and addition is jointly continuous, but scalar multiplication is only jointly continuous once one restricts the scalar to be non-negative.

Below the fold, I record the textbook proof of Theorem 2 and Theorem 1. There is nothing particularly original in this presentation, but I wanted to record it here for my own future reference, and perhaps these results will also be of interest to some other readers.

— 1. Proof of Theorem 2 —

Let V be a finite-dimensional Hausdorff topological vector space, with topology $\mathcal{F}$. We need to show that every set which is open in the usual topology, is open in $\mathcal{F}$, and conversely.

Let $v_1, ..., v_n$ be a basis for the finite-dimensional space V. From the continuity of the vector space operations, we easily verify that the linear map $T : \mathbf{R}^n \to V$ given by

$$T(x_1, ..., x_n) := x_1 v_1 + ... + x_n v_n$$

is continuous. From this, we see that any set which is open in $\mathcal{F}$, is also open in the usual topology.

Now we show conversely that every set which is open in the usual topology, is open in $\mathcal{F}$. It suffices to show that there is a bounded open neighbourhood of the origin in $\mathcal{F}$, since one can then translate and dilate this open neighbourhood to obtain a (sub-)base for the usual topology. (Here, "bounded" refers to the usual sense of the term, for instance with respect to an arbitrarily selected norm on V (note that on a finite-dimensional space, all norms are equivalent).)

We use T to identify V (as a vector space) with $\mathbf{R}^n$. As T is continuous, every set which is compact in the usual topology, is compact in $\mathcal{F}$. In particular, the unit sphere $S^{n-1} := \{x \in \mathbf{R}^n : \|x\| = 1\}$ (in, say, the Euclidean norm $\| \|$ on $\mathbf{R}^n$) is compact in $\mathcal{F}$. Using this and the Hausdorff assumption on $\mathcal{F}$, we can find an open neighbourhood U of the origin in F which is disjoint from S^{n-1}.

At present, U need not be bounded (note that we are not assuming V to be locally connected *a priori*). However, we can fix this as follows. Using the joint continuity of the scalar multiplication map, one can find another open neighbourhood U' of the origin and an open interval $(-\epsilon, \epsilon)$ around 0 such that the product set $(-\epsilon, \epsilon) \cdot U' := \{tx : t \in (-\epsilon, \epsilon); x \in U'\}$ is contained in U. Since U avoids the unit sphere S^{n-1}, U' must avoid the region $\{x \in \mathbf{R}^n : \|x\| > 1/\epsilon\}$ and is thus bounded, as required.

> **Corollary 3** In a Hausdorff topological vector space V, every finite-dimensional subspace W is closed.

Proof: It suffices to show that every vector $x \in V \setminus W$ is in the exterior of W. But this follows from Theorem 2 after restricting to the finite-dimensional space spanned by W and x. □

— 2. Proof of Theorem 1 —

Let V be a locally compact Hausdorff space, thus there exists a compact neighbourhood K of the origin. Then the dilate $\frac{1}{2}K$ is also a neighbourhood of the origin, and so by compactness K can be covered by finitely many translates of $\frac{1}{2}K$, thus

$$K \subset S + \frac{1}{2}K$$

for some finite set S. If we let W be the finite-dimensional vector space generated by S, we conclude that

$$K \subset W + \frac{1}{2}K.$$

Iterating this we have

$$K \subset W + 2^{-n}K$$

for any $n \geq 1$. On the other hand, if U is a neighbourhood of the origin, then for every $x \in V$ we see that $2^{-n}x \in U$ for sufficiently large n. By compactness of K (and continuity of the scalar multiplication map at zero), we conclude that $2^{-n}K \subset U$ for some sufficiently large n, and thus

$$K \subset W + U$$

for any neighbourhood U of the origin; thus K is in the closure of W. By Corollary 3, we conclude that

$$K \subset W.$$

But K is a neighbourhood of the origin, thus for every $x \in V$ we have $2^{-n}x \in K$ for all sufficiently large n, and thus $x \in 2^n W = W$. Thus $V = W$, and the claim follows.

Comments (20)



JuanPi · 24 May 2011

Hi,

Thanks for the round-up. I am reading . Khatskevich, D.Shoikhet, Differentiable Operators and Nonlinear Equations so this is a nice backup.

Now, I have a more or less green-horn question: If we have D a nonlinear differential operator acting on some Hilbert space of functions. Is there anything we can tell about the set of all solutions q of the problem D(q) = 0? In the linear case we know they are vector space (the nullspace), what can be said in the nonlinear case?

Sixth Linkfest · 24 May 2011

[...] Terence Tao: Stein's maximal principle, Stein's spherical maximal theorem, Locally compact topological vector spaces [...]



MysticTerminator · 24 May 2011

How is R in the cocompact topology Hausdorff?

[Ah, yes, it isn't – thanks – T.]



Allen Knutson · 24 May 2011

Theorem 2, as presently stated, is a bit too strong.

[Corrected, thanks – T.]



Anonymous Rex · 25 May 2011

Hi, tiny typo: In "Combining the two statements, we see that for every {x \in U"}, we have {sx \in U} for every {x \in (-\epsilon,\epsilon)}", you use x twice. I assume you mean for s to be the scalar.

[Corrected, thanks – T.]

Anonymous · 31 May 2011

Dear Professor Tao,
Grothendieck (MR0372565 (51 #8772)) attributes Theorem 1 to Banach, but
the oldest reference I could find was the 1953 edition of Bourbaki's TVS
(MR0910295 (88g:46002)), so Weil may have done it first after all.

Hilbert's fifth problem and Gleason metrics « What's new · 17 June 2011

[...] space, it must be isomorphic to a finite-dimensional vector space with the usual topology (see this blog post for a [...]

wildildildlife · 4 February 2012

In the first sentence below the head '1. Proof of Theorem 2 ', and in Corollary 3, you write 'topological space' which should be 'topological vector space'.
[Corrected, thanks – T.]

Richard Hevener · 7 May 2012

Thanks for a very nice treatment of these results.
I believe the last part of your proof of Thm. 2 can be simplified quite a bit: Given your U, by the continuity of scalar mult., we can find a nbhd. U' of the origin and a nbhd. N = (-e,e) of 0 such that if x is in U' and s is in N, then sx is in U. As at the end of your argument, since U avoids the unit sphere, U' must be contained in the 1/e ball.
[Simplification incorporated, thanks – T.]

Richard Hevener · 7 May 2012

Inside the brackets, you should say (-e,e) instead of (1-e,1+e).
[Corrected, thanks – T.]

Richard Hevener · 7 May 2012

One more typo: again inside the braces, it should be (-e,e) instead of (e,e).
[Gah! Corrected -T.]

Number of group topologies on the real line | Mathematical Notes · 7 September 2013

[...] locally compact topological vector space is finite-dimensional (the reader might see Tao's article on his blog or Bourbaki's book, Topological vector spaces). Therefore, the weak topologies [...]

Kasper Henriksen · 5 April 2015

I think I've found a slightly different argument that doesn't require introducing an open neighbourhood U of 0. If $\dim V = 0$, there is obviously nothing to prove. Otherwise we can write V as the union of its one-dimensional subspaces ("L" is for lines):

$$\mathcal{L} = \{L \subseteq V : L \leq V, \dim L = 1\}.$$

Then
\[\bigcap_n 2^{-n}K
= \bigcup_{L \in \mathcal{L}} \left(\bigcap_n 2^{-n}K \cap L\right)
= \bigcup_{L \in \mathcal{L}} \{0\}
= \{0\}
\]
by compactness of $K \cap L$ and the fact that $L \cong \mathbb F$.

Anonymous · 11 October 2015

> in an infinite-dimensional normed vector space V with the strong topology, an iteration of the Riesz lemma shows that the closed unit ball B in that space contains an infinite sequence with no convergent subsequence, which (*by the Heine-Borel theorem*) implies that V cannot be locally compact.

Would you elaborate how the Heine-Borel theorem (which says that in a metric space a subset is compact if and only if it is complete and totally bounded) is used here?



Terence Tao · 11 October 2015

The formulation of the Heine-Borel theorem I am using here is that for a metric space X, the properties of (a) being topologically compact (every open cover has a finite subcover), (b) sequentially compact (every sequence has a convergent subsequence), and (c) being complete and totally bounded, are equivalent. For this particular application only the equivalence of (a) and (b) is needed.

Anonymous · 11 October 2015

What does the "usual topology" mean in the statement of Theorem 2? Let (X, τ) be a hausdorff TVS (over $\mathbf{R}$) of finite dimension d. Do you mean that τ must be the standard topology on $\mathbf{R}^d$, which means you are *identifying* X with $\mathbf{R}^d$?



Terence Tao · 11 October 2015

Yes. Or more precisely, one selects a linear isomorphism between X and $\mathbf{R}^d$, and then uses this isomorphism to pull back the usual topology on $\mathbf{R}^d$. This is independent of the choice of isomorphism, because all linear maps from $\mathbf{R}^d$ to itself are automatically continuous in the standard topology.



Adam · 9 June 2019

In the proof of Theorem 1, why does $2^{-n}K \subset U$ for some large enough n? From compactness there exists a finite set F such that, $K \subset \bigcup_{i \in F} 2^i U$. It seems for me that the (non-obvious, and probably not true) fact that for $i < j$, $2^i U \subset 2^j U$, was used.



Adam · 9 June 2019

I've looked up more. I found that we can assume without loss of generality that U is a balanced set, which fixes every problem I had earlier.



Jochen Wengenroth · 13 February 2023

Theorem 2 is stated and proved in A. Tychonoff, Ein Fixpunktsatz, Math. Ann. 111, 767–776 (1935).